

RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year 2020 – 2021 (Odd Semester)

Innovative practice(s) description

Course code and Title: ME8793 Process Planning and Cost Estimation

Name of the Instructor(s): Dr.S. Rajakarunakaran, Prof/Mech, Mr.R.Venkatesh, AP/Mech,

Date & Time: 13.12.2020 & 3.30 PM to 4.30 PM

Theory to Practical (T2P)

Topic: Production Cost Estimation & Machining Time Calculation

Product Costing & Machining Time is one of the key areas in Manufacturing and process industries. It is used for estimating and valuating the internal cost of a product.

Initially we taught unit IV & Unit V Production Cost Estimation and Machining Time Calculation during class hour. After completion of Unit IV & V, Students were divided into several groups and each group to calculate,

- Cost of different welding process and (Butt Joint & Lap Joint) at our workshops.
- Machining Time for Various Manufacturing Processes at our Manufacturing Technology Laboratory.

Production Cost Estimation



Machining Time Calculation



Outcome:

- This theory to practice was important to attain practical hands-on experience.
- Students understand the importance of costing and estimation.
- Students estimate the cost of different types of job such as welding shop and foundry shop.

This activity (Theory to Practical) helps in attaining the following CO – PO mapping:

<u>Course Outcome / Programme Outcomes</u>	<u>PO1</u>	<u>PO2</u>	<u>PO3</u>	<u>PO4</u>	<u>PO5</u>	<u>PO8</u>	<u>PO9</u>	<u>PO10</u>	<u>PO12</u>
CO4 - Compute the job order cost for different type of shop floor.	2	1	2	1	1	2	1	1	3
CO5 – Calculate the machining time for various machining operations.	2	3	3	2	2	2	1	2	3